SARA

Smart Assistive Reading Aid

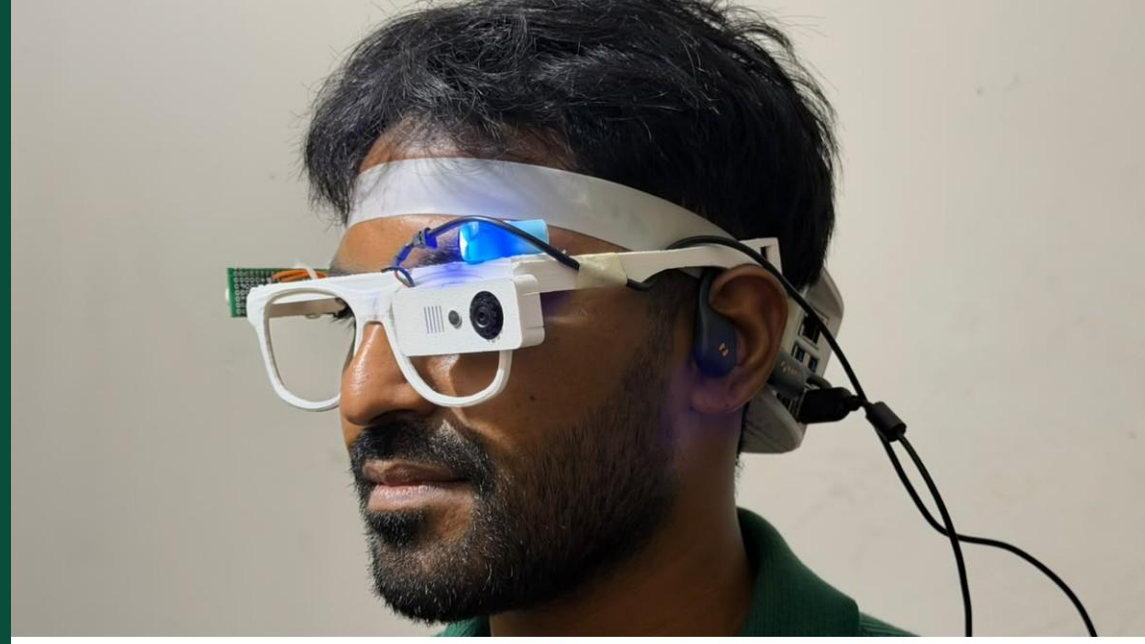
Offline AI Smart Glasses for Reading, Object Recognition and Bangladeshi Taka Verification

MD Abdun Fattah Lam (2220014) · Mukut Protim Memo (2020086)

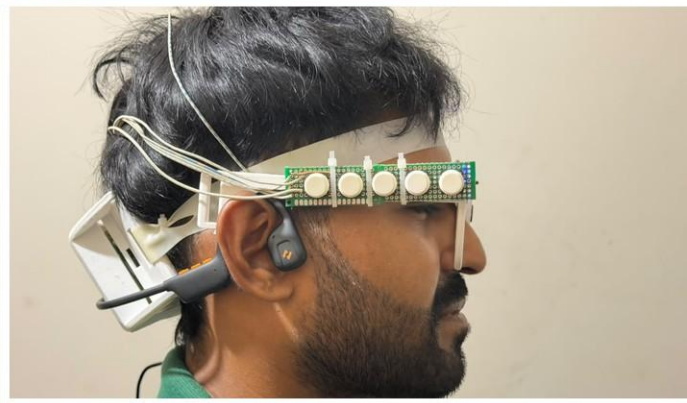
Tanjila Hossain Tapti (2130561)

Supervisor: Mohammad Shidujaman, PhD, Assistant Professor

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Three everyday challenges

Money

Reading

Surroundings







Illustration

Why existing solutions fall short

2.2 billion

people worldwide have a near or distance vision impairment [1]

most of them in low- and middle-income regions [1]



Limited Bangla support

OrCam MyEye does not support Bangla [2]



Internet dependence

Research glasses send pictures to a server [3, 4]



Limited Taka verification

Reviewed systems do not name Taka notes [2, 3, 4]

Sources: [1] WHO 2023; [2] OrCam MyEye; [3] Ali A. et al., IEEE Access 2021; [4] Brilli et al., arXiv 2024.

SARA in action: ask the glass, not a stranger



Our prototype

Speak
"Glass, check money"

→

Detect offline
Raspberry Pi 5, no internet

→

Hear the answer
Bangla speech + vibration

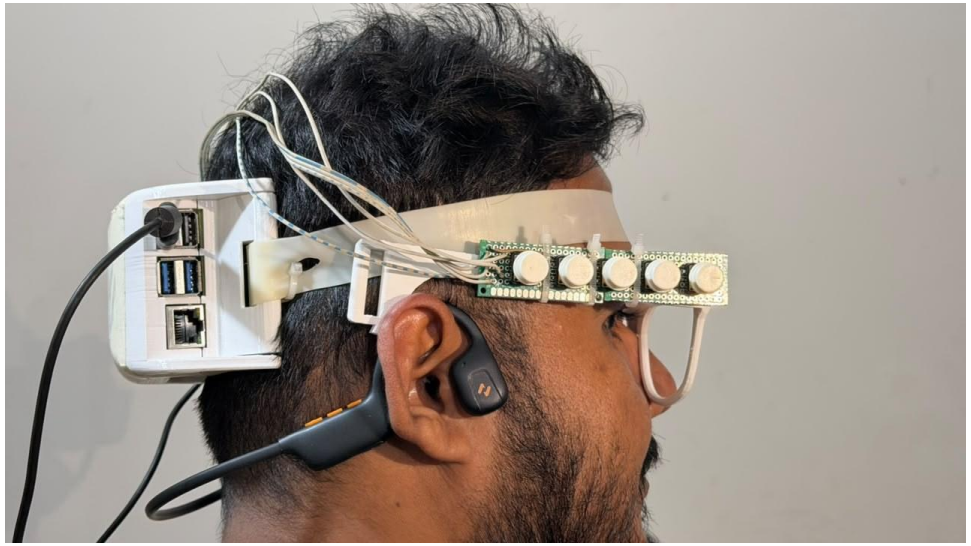
Money
Names the note and screens it with care

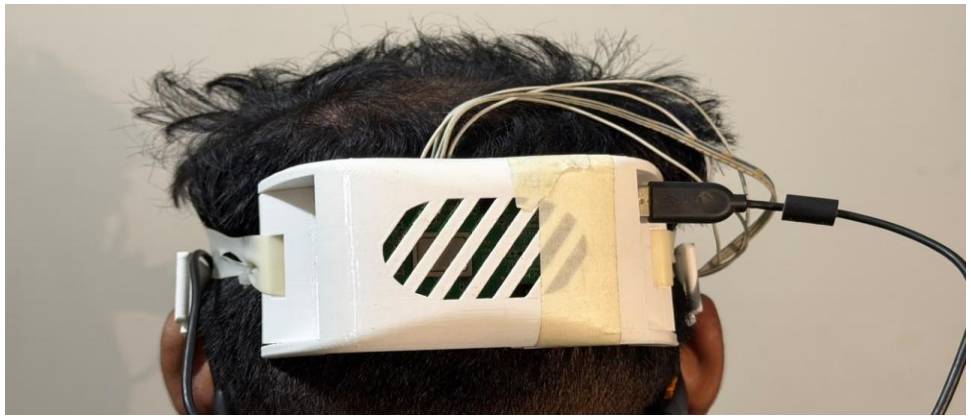
Reading
Reads Bangla and English labels aloud

Surroundings
Names people and objects ahead

পাঁচশত টাকার নোট "500 Taka note": the answer the glass speaks

Five modes, started by button or by voice





Text reading

Bangla and English

Objects

Named in Bangla

Taka notes

Value + safe check

Emotion

Face in front

Online description

Optional; needs internet

How the user starts a mode: five buttons on the frame, or a voice command: "Glass, read text"

Six research questions

1 System

Can it all run on a Raspberry Pi 5?

2 Synthetic data

Can unlabelled training data detect real notes?

3 Number of views

Can one view of a note be enough?

4 Unseen prints

Does the check hold on new counterfeit prints?

5 Safety

Can it never accuse a genuine note?

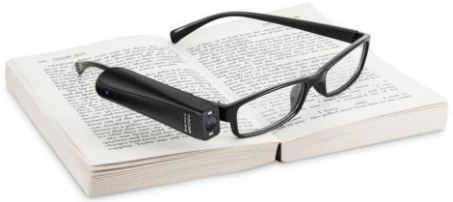
6 Bangla OCR

What matters most for reading Bangla?

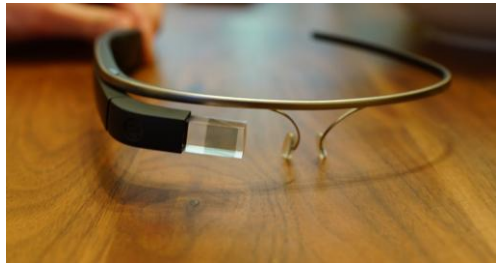
What existing devices miss

System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	Yes	Yes
Google Glass scene analysis [3]	No	No	No	Yes
AIris (2024) [4]	No	No	No	Yes
Raspberry Pi reading aids *	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

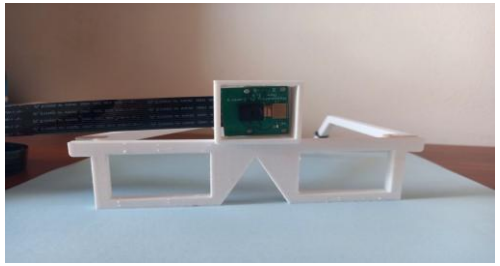
*Rows with a reference: from the cited source. * Our assessment of typical hobby builds, not a cited result.*



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

The research gap

None of the systems we reviewed combines Bangla, Taka and offline use.

- Counterfeit tests reuse the same prints
- Few report speed on the real device

Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

How the glass works

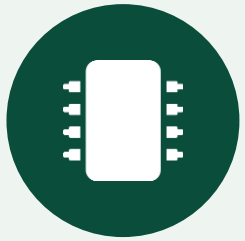


Camera, buttons, voice

Camera 640×480

5 buttons on the frame

Voice: "glass, …"



Raspberry Pi 5

Mode manager

All models on the CPU

Offline, except the online mode



AI modules

Text: CRAFT + EasyOCR [5, 6]

Objects: YOLOv8s, COCO [7, 8]

Taka: YOLOv8s + safe checks

Emotion: YuNet + EfficientNet [9]

Online: cloud model (optional)



Audio + vibration

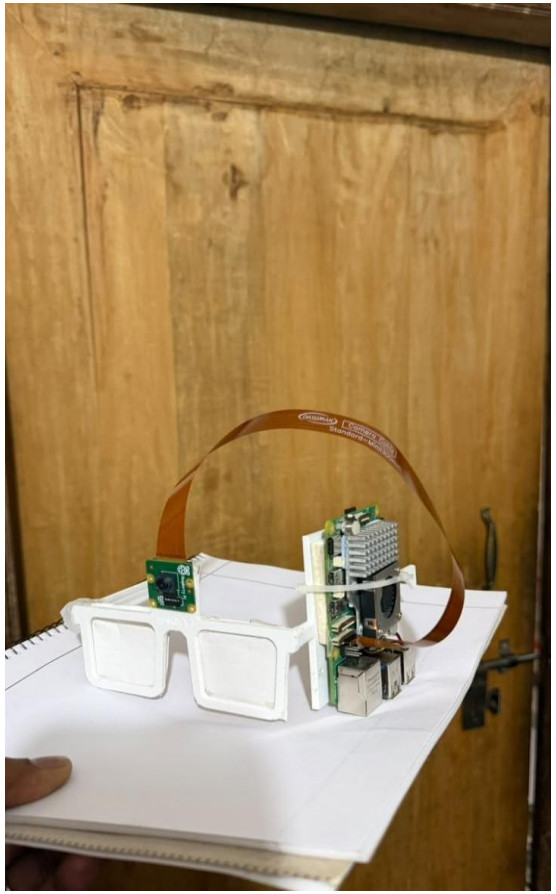
Bangla or English speech [10, 11]

Vibration for a note

Everything runs on the glass itself; only the optional online mode uses the internet.

Sources: [5] Baek et al. 2019; [6] EasyOCR; [7] Ultralytics YOLOv8; [8] COCO; [9] RAF-DB; [10] Piper; [11] eSpeak NG.

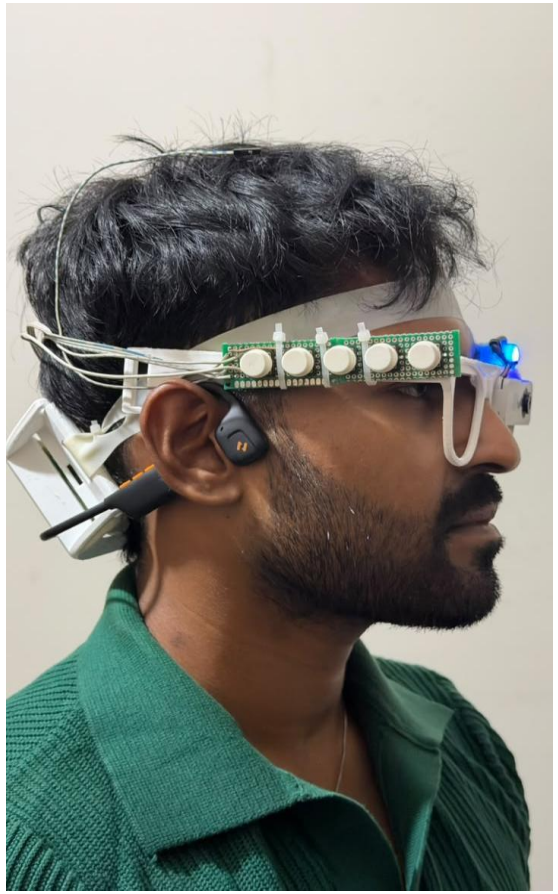
From a desk prototype to a wearable glass



Version 1
Board on one temple



Front
Camera + UV lamp



Side
5 buttons, bone-conduction earphone



Rear
Pi 5 in a ventilated case

Taka detection: 22,333 training images, no hand labels



Note photos [12] are pasted onto real scenes [8], and the program writes every label itself.

SYNTHETIC COMPOSITES

0.995

mAP@0.5 on 2,282 held-out composites

REAL PHOTOS

94.1%

right note value on 1,536 photos taken by other people

Sources: [12] BanglaTaka (Mendeley); [8] Lin et al., COCO, ECCV 2014; detector [7] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Notes held in a hand: from 1 in 8 to 1 in 2

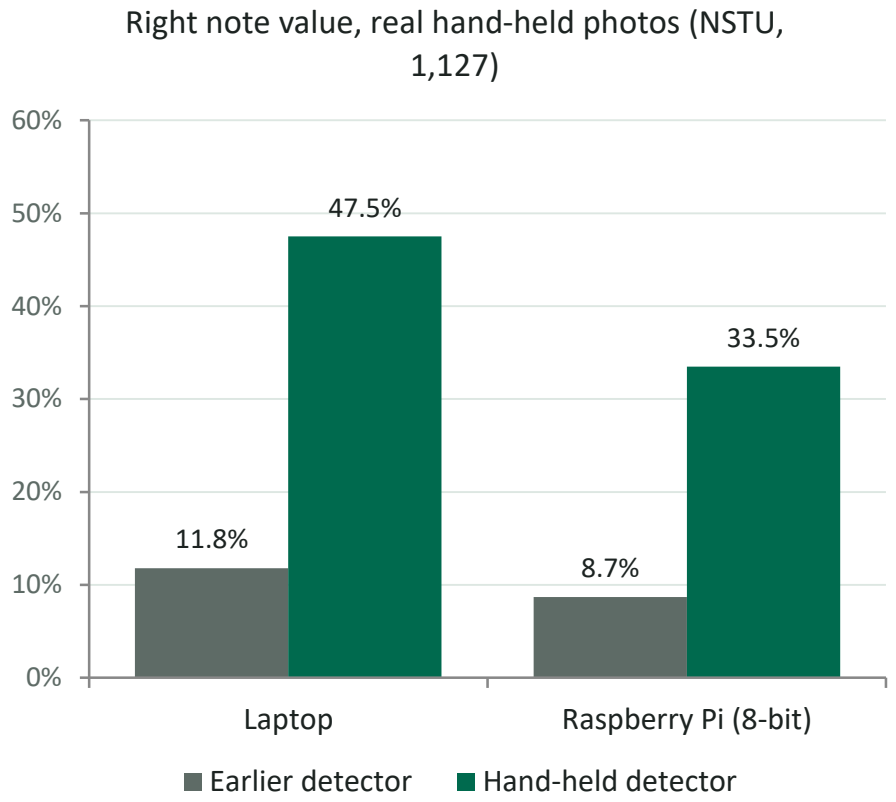
TRAINING: 6,000 SYNTHETIC CLOSE-UPS



TEST: REAL HAND-HELD PHOTOS

11.8% → 47.5%

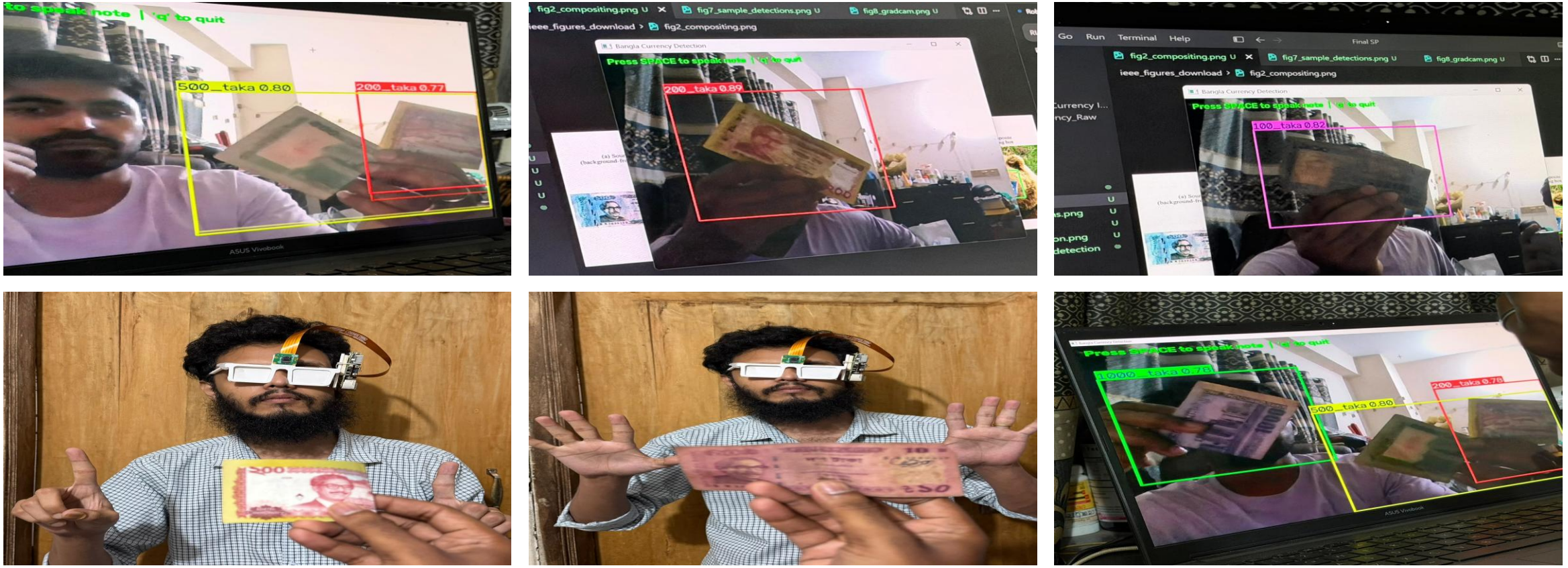
Right note value on the laptop. Trade-off: with 3 notes in view, the total is right 97.0% (was 99.2%).



Sources: test photos [13] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [7] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

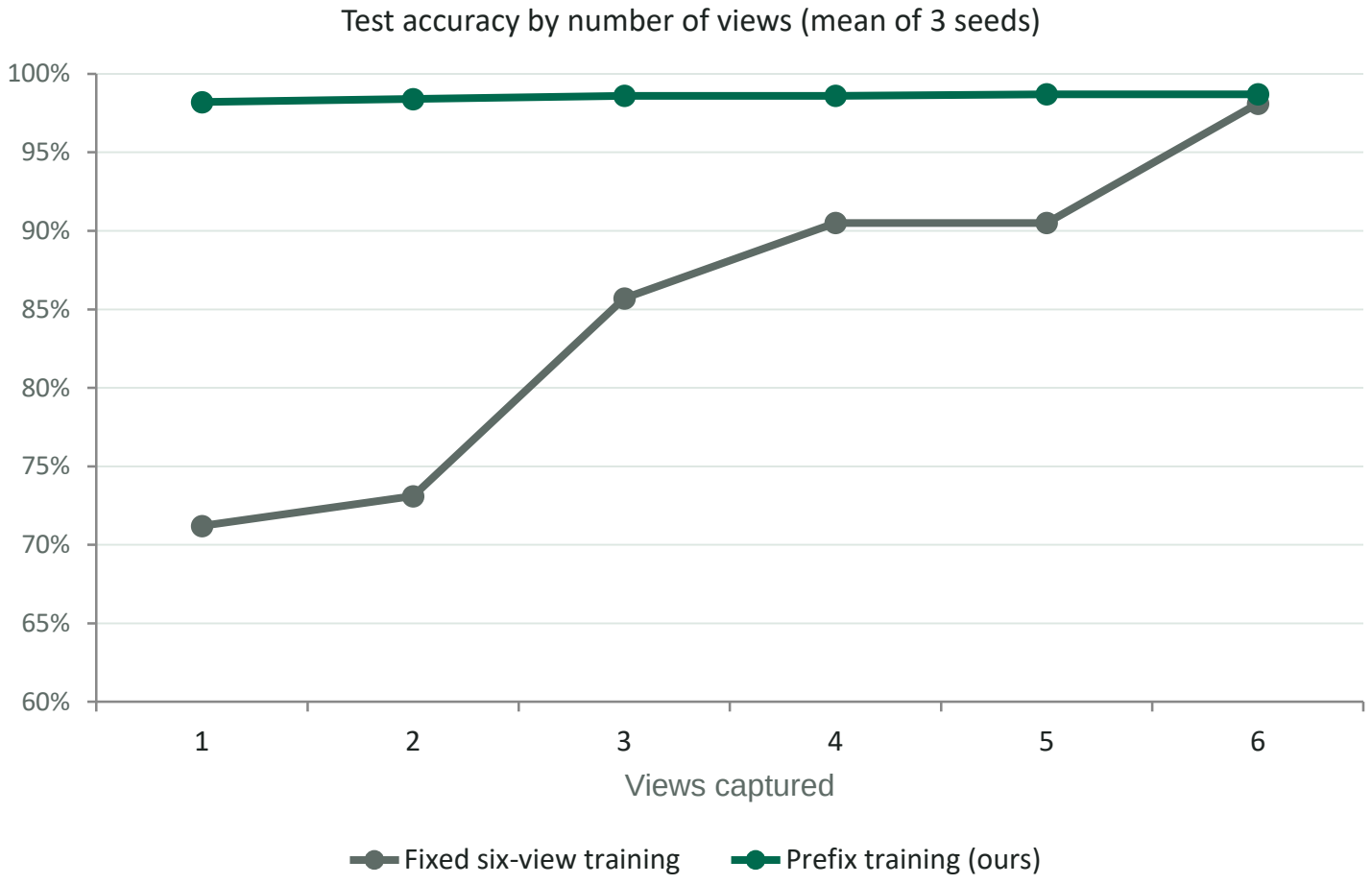
Live tests with real notes

Informal demonstrations, not a measured accuracy study



Six note values (10 to 1,000 Taka) named at 0.63–0.89 confidence, including worn and folded notes. Laptop webcam and the first glass prototype.

Counterfeit check: one view is enough with prefix training



With a single view

71.2%

rises to

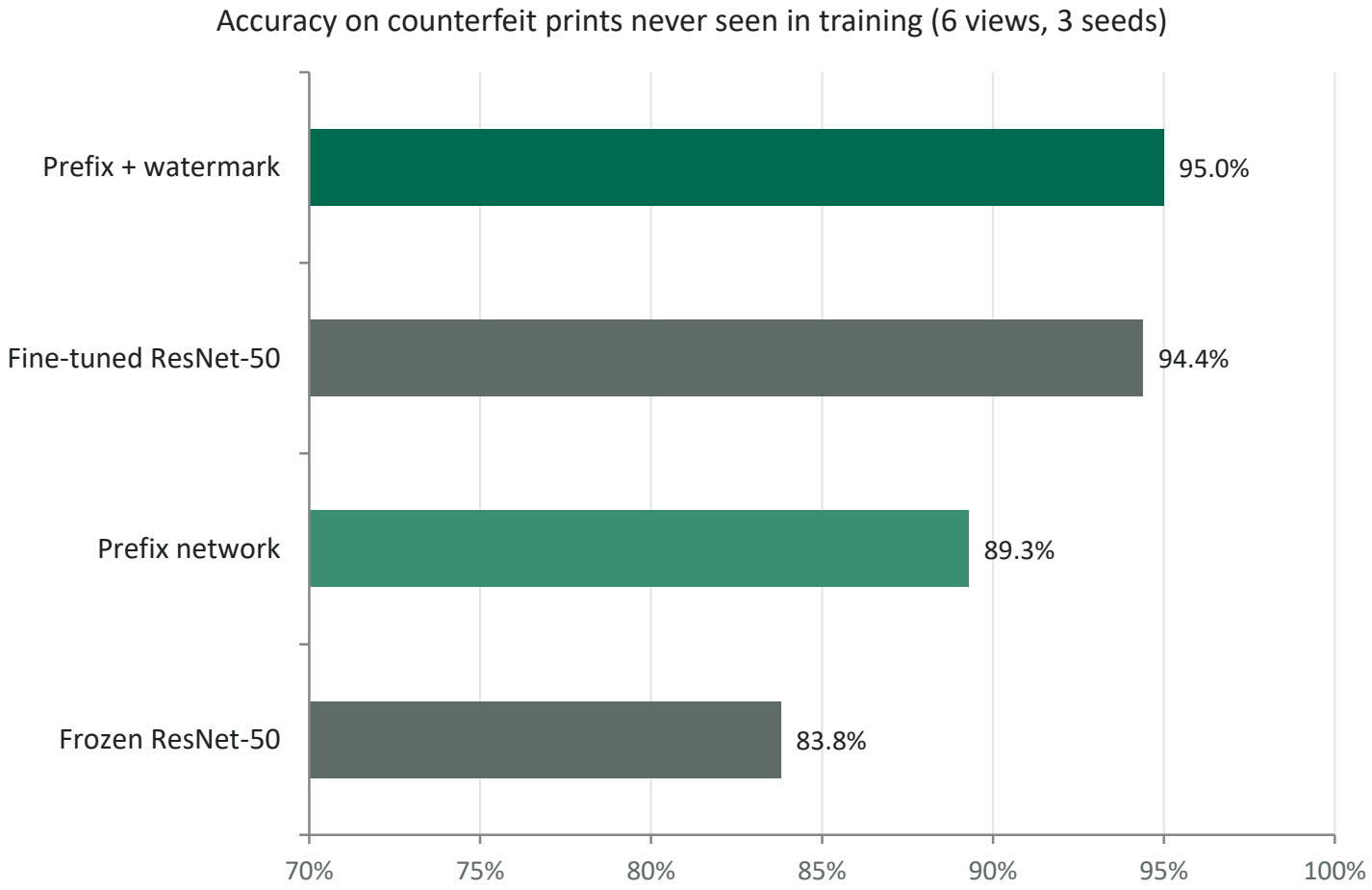
98.2%

Training on random first views: one view is enough.

Held-out JaalTaka test notes, never used for training or for any choice

Data: [14, 15] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Unseen counterfeit prints: a stricter test



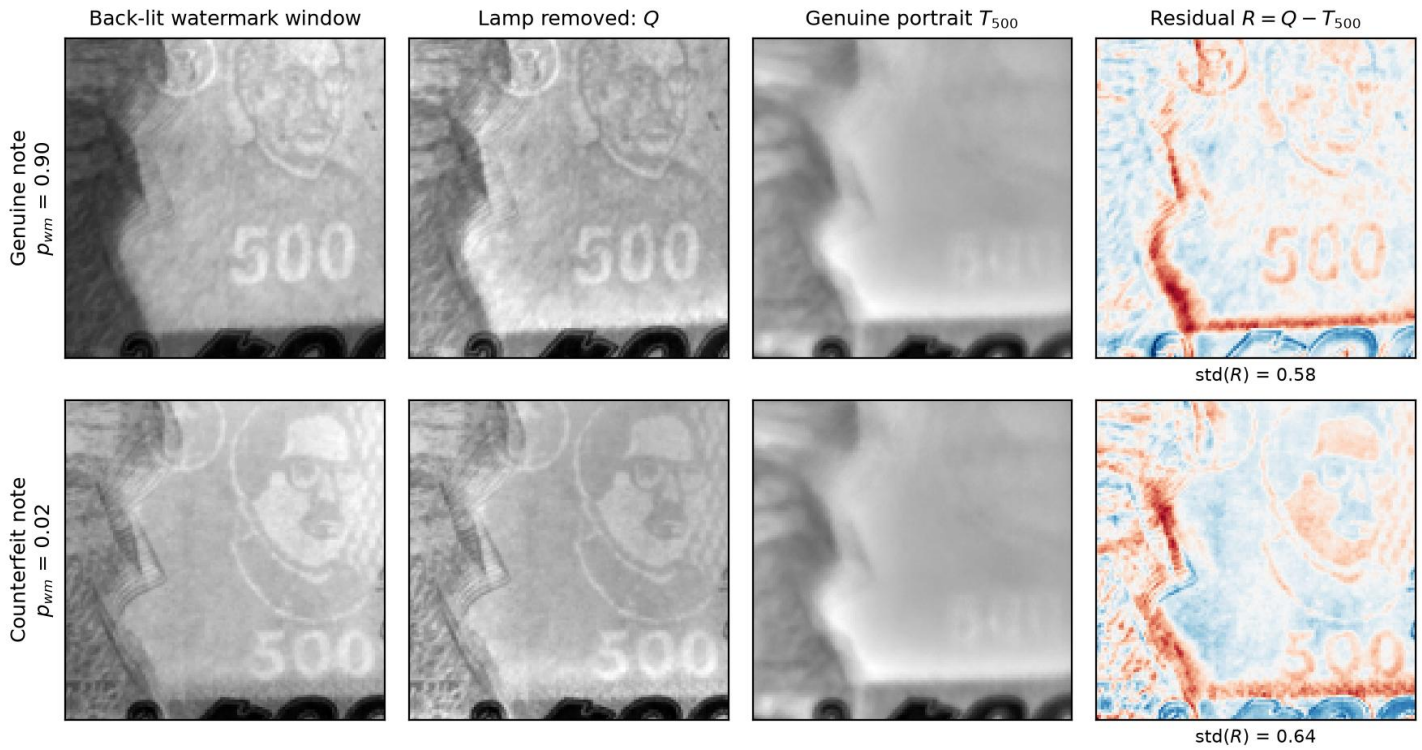
95.0%

on counterfeit prints never seen in training, with the watermark check

Why a stricter split: 279 of 322 readable 500 Taka counterfeits share one serial number, so the usual split tests copies of known prints.

Data: [14] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [16] SVM on security features; [17] CNNs; [19] UV flecks + RPN.

DMWR: a tiny, physics-based watermark checker



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

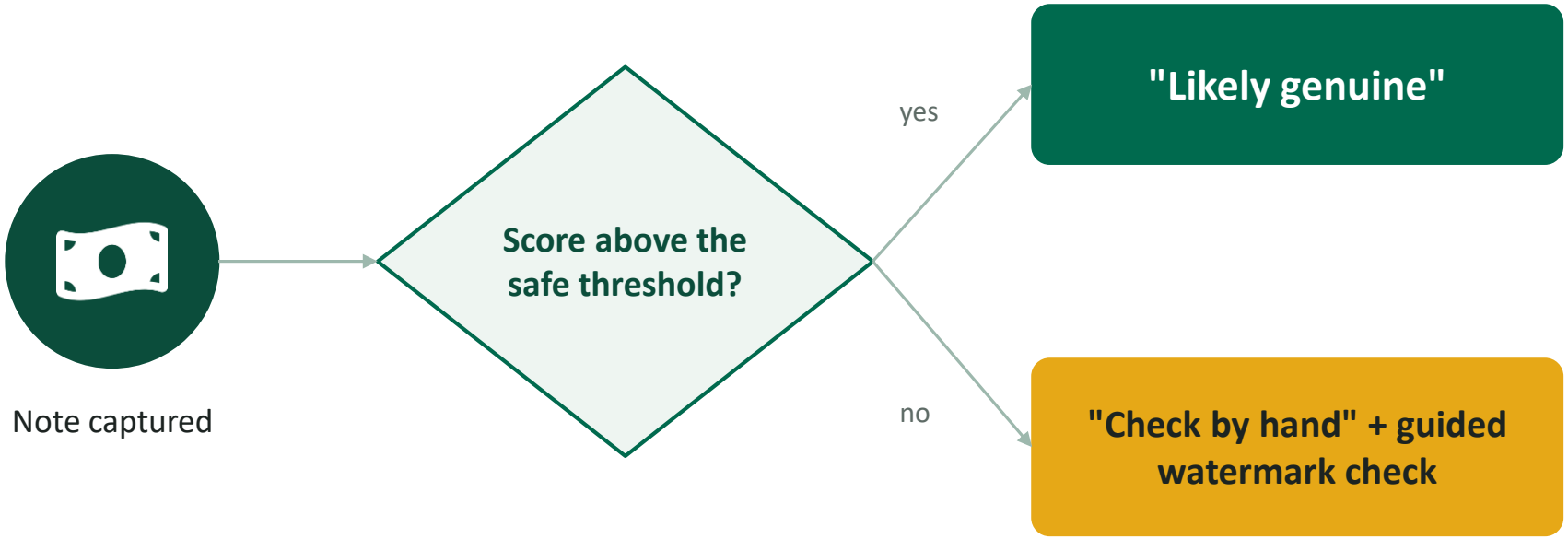
on unseen counterfeit prints (3 seeds, ±0.3)

46× smaller

than MobileNetV2 [18] (47,818 parameters, 0.21 MB), and as accurate: 92.9%, p ≥ 0.375

Baseline [18] Sandler et al., MobileNetV2, CVPR 2018; data [14] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe before decisive: the glass never says "counterfeit"



0 / 88

test counterfeits passed as "likely genuine"

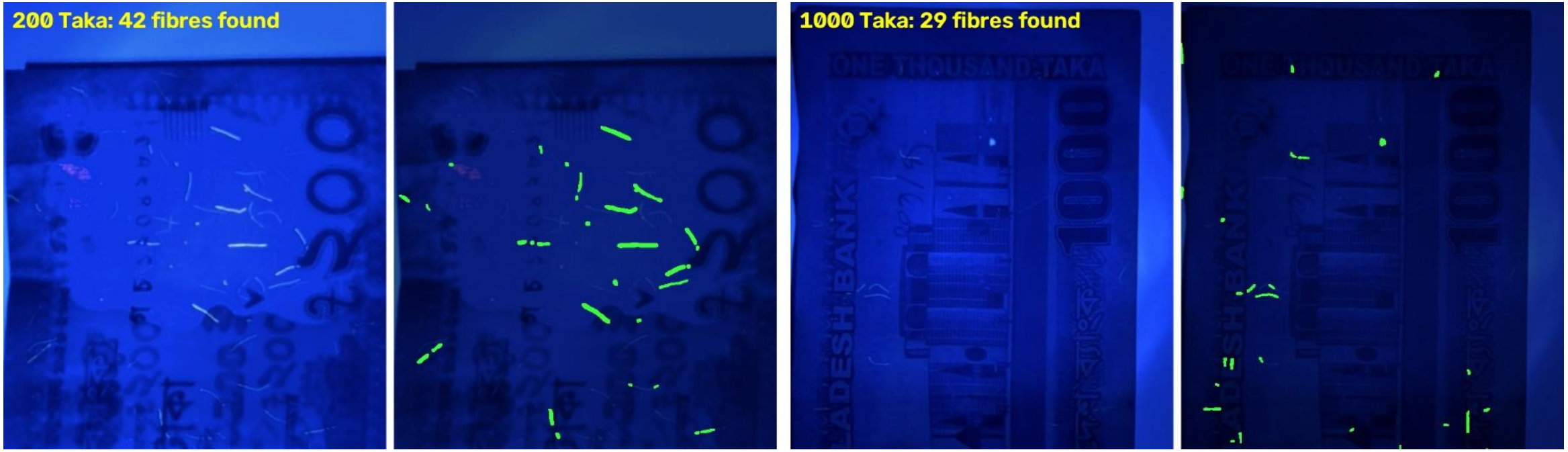
"Likely genuine": the score is higher than every counterfeit seen in validation (57.5% of genuine notes).

"Check by hand": the glass is not sure. It never means "counterfeit".

A screening aid, not a guaranteed counterfeit detector

Test notes: [14] JaalTaka.

UV fibre check: the glass carries its own UV lamp



85.6%

of 111 real UV photos of genuine notes accepted

88.6%

of SIMULATED fibre-free sheets rejected (not real counterfeits)

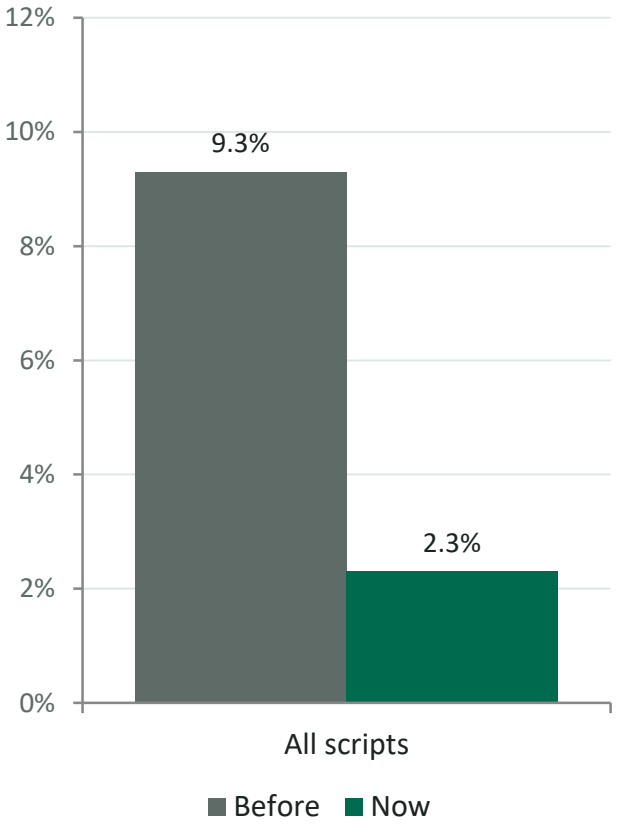
Prototype, off by default

The rejection result is simulated; it must be validated on real counterfeit notes

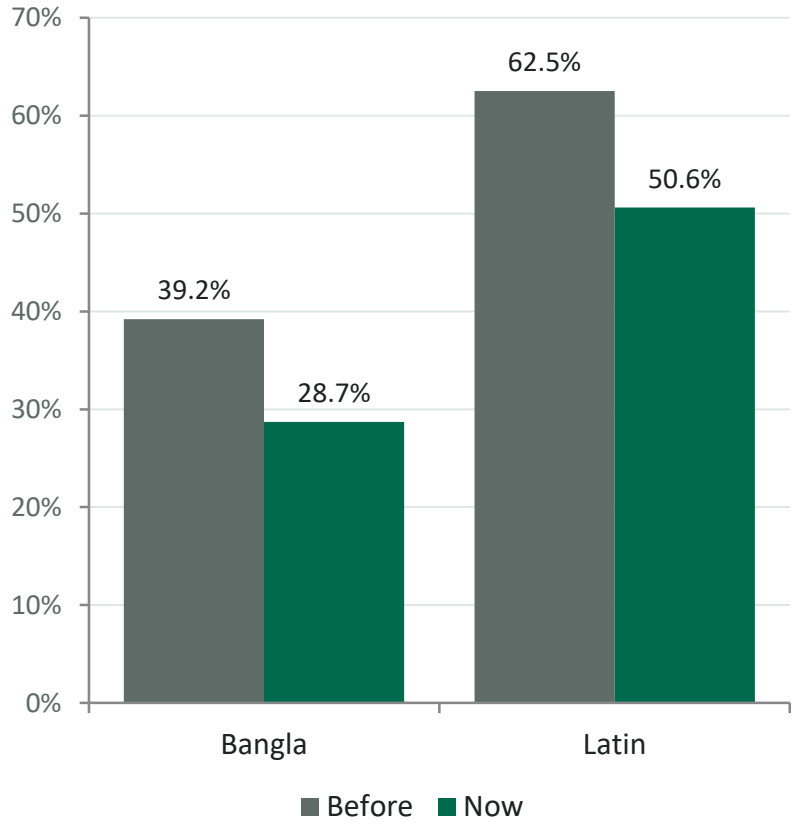
The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [20]. Prior Taka UV work: [19] (96.3%). Fibre check and photos are ours.

Bangla reading: rendered text and real street photos

Rendered text (clean)



Real word photos (hard)



What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words

What did not

Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

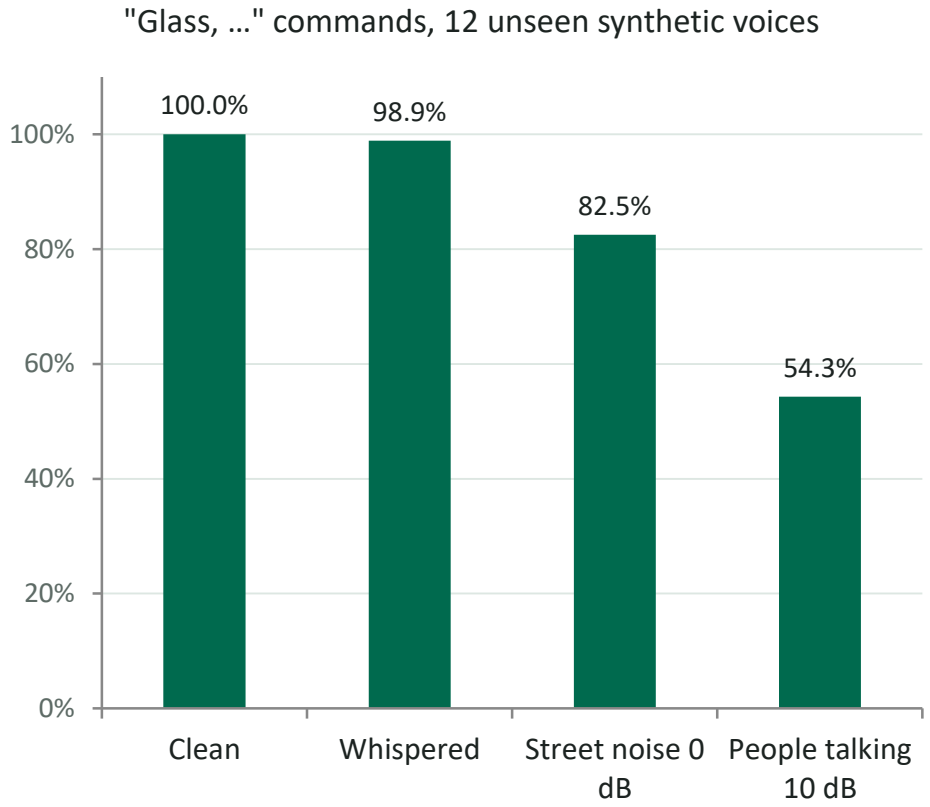
Character error rate, lower is better. Measured on evaluation datasets, not yet through the glass.

Engine [6] EasyOCR with [5] CRAFT; real words from [21] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Voice: a natural Bangla voice and spoken commands

Speaking

- Natural Piper voice, Bangla and English [10, 11]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Command results: synthetic voices on a laptop, not human trials on the glasses

Recogniser [22] Vosk; noise from [23] Piczak, ESC-50; voices [10, 11]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Running on the Raspberry Pi 5

399 ms
currency check, end to end

64 ms
watermark check

562 ms
object scan (YOLOv8s)

680 ms
emotion, face in view

18.3 s
3-line sign, text found once

1.48 GB
peak memory, 30-minute run



Key finding: heat

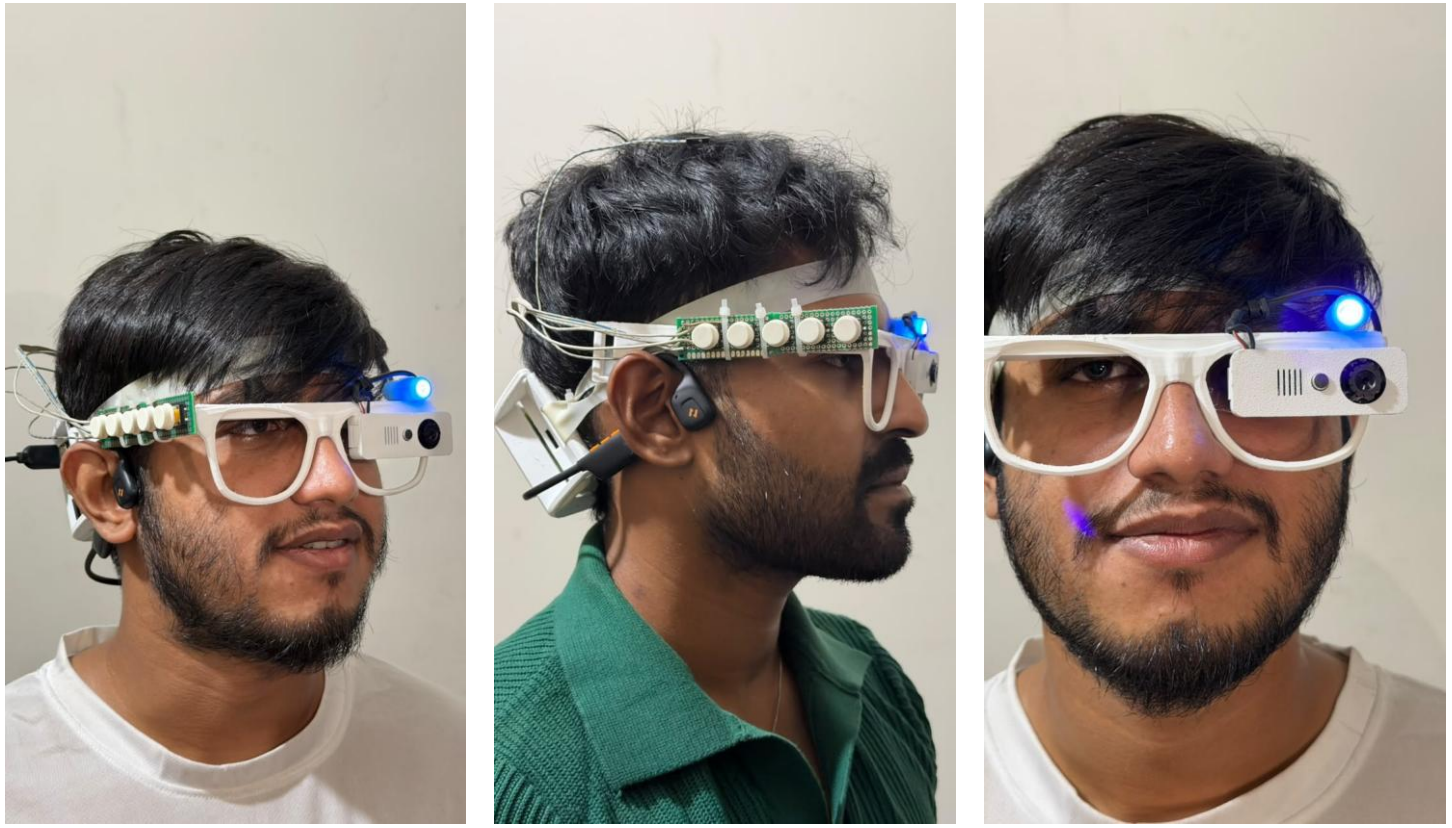
88.7 °C

peak temperature; the Pi throttled in 76% of samples (idle 71 °C)

Response: a thermal governor slows background scans when hot; better cooling is future work.

Measured on a Raspberry Pi 5 (8 GB) on 13 October 2026, board at its temperature limit; memory from the 1 October run. Changes after 13 October are not yet timed on the Pi.

Testing the worn glass: where the project stands



Prototype version 2 on team members: fit, weight and button reach are checked before user trials.

DONE

Engineering tests

Each part tested on held-out data, on a laptop and the Pi 5

IN PROGRESS

Participant trials

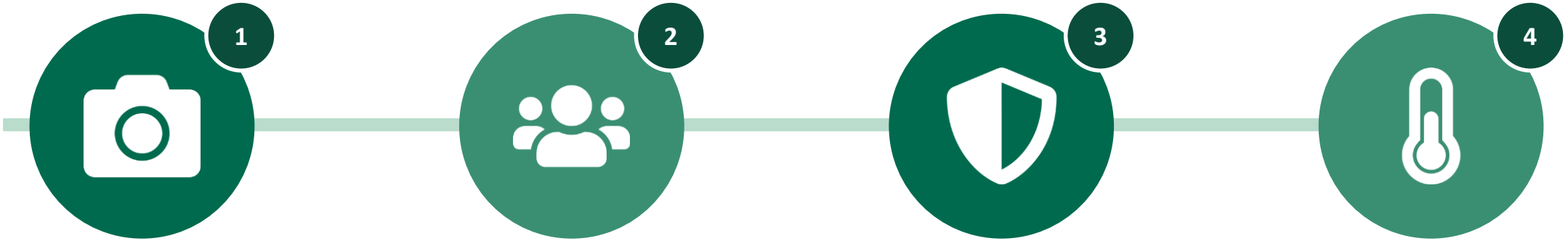
Sessions with a visually impaired participant; no user study complete yet

LIMITATION

Glass camera

First field test: currency unreliable, Bangla text not read (640×480, fixed focus)

Future work: the roadmap



Camera

Higher resolution or autofocus

User study

8–12 participants, including their own voice commands

More banknotes

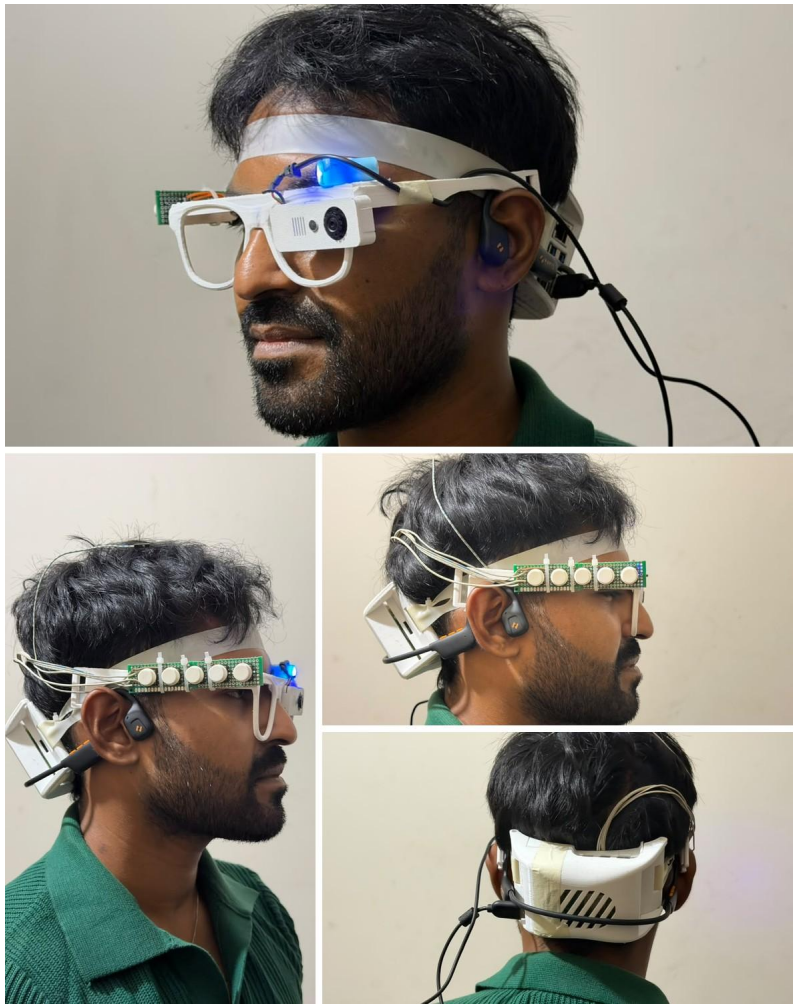
Unseen counterfeit prints; UV check on real fakes

Speed and cooling

Time the new code on the Pi; better cooling; battery

Done: engineering tests **In progress:** participant trials **Future:** the four steps above

Conclusion



Built

An offline glass that speaks Bangla, reads, names objects and Taka on a Raspberry Pi 5

New methods

Prefix training, a safe verdict, the DMWR watermark check, hand-held fine-tuning

Next

A better camera and the user trials

Thank you

Questions are welcome



References

[1] World Health Organization. Blindness and vision impairment: fact sheet, 2023.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[6] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[7] Jocher, G., Chaurasia, A., Qiu, J. Ultralytics YOLOv8.

[8] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[9] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[10] Rhasspy. Piper: a fast, local neural text-to-speech system.

[11] eSpeak NG text-to-speech.

[12] A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data.

[13] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[14] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[15] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[16] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[17] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[18] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[19] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[20] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[21] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[22] Alpha Cephei. Vosk offline speech recognition toolkit.

[23] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment